

```
mysql> use cinesphere;  
Reading table information for completion of table and column names  
You can turn off this feature to get a quicker startup with -A
```

Database changed

```
mysql> show tables;
```

```
+-----+
```

```
| Tables_in_cinesphere |
```

```
+-----+
```

```
| booking              |
```

```
| booking_seat         |
```

```
| customer             |
```

```
| genre                |
```

```
| movie                |
```

```
| movie_genre          |
```

```
| payment              |
```

```
| screen               |
```

```
| screening            |
```

```
| seat                 |
```

```
| slot                 |
```

```
| theater              |
```

```
| ticket               |
```

```
+-----+
```

```
13 rows in set (0.001 sec)
```

```
mysql>
```

```
+-----+-----+-----+-----+-----+-----+
| Table | Create Table
|
+-----+-----+-----+-----+
| movie | CREATE TABLE `movie` (
| `movie_id` int NOT NULL AUTO_INCREMENT,
| `title` varchar(200) NOT NULL,
| `duration_minutes` int NOT NULL,
| `language` varchar(50) NOT NULL,
| `release_date` date DEFAULT NULL,
| `certificate` varchar(10) DEFAULT NULL,
| `rating` decimal(3,1) DEFAULT NULL,
| `synopsis` text,
| PRIMARY KEY (`movie_id`),
| CONSTRAINT `chk_movie_duration` CHECK ((`duration_minutes` > 0)),
| CONSTRAINT `chk_movie_rating` CHECK (((`rating` >= 0) and (`rating` <= 10)))
| ) ENGINE=InnoDB AUTO_INCREMENT=5 DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+-----+-----+-----+-----+
1 row in set (0.001 sec)
```

100

```
mysql> SHOW CREATE TABLE genre;
```

```
+-----+-----+
| Table | Create Table                                     |
+-----+-----+
| genre | CREATE TABLE `genre` (
  `genre_id` int NOT NULL AUTO_INCREMENT,
  `genre_name` varchar(50) NOT NULL,
  PRIMARY KEY (`genre_id`),
  UNIQUE KEY `uq_genre_name` (`genre_name`)
) ENGINE=InnoDB AUTO_INCREMENT=11 DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+-----+
1 row in set (0.001 sec)
```

```
mysql> █
```

```
mysql> SELECT
->     TABLE_NAME,
->     COLUMN_NAME,
->     CONSTRAINT_NAME,
->     REFERENCED_TABLE_NAME,
->     REFERENCED_COLUMN_NAME
-> FROM information_schema.KEY_COLUMN_USAGE
-> WHERE TABLE_SCHEMA = 'cinesphere'
->     AND REFERENCED_TABLE_NAME IS NOT NULL
-> ORDER BY TABLE_NAME, COLUMN_NAME;
```

TABLE_NAME	COLUMN_NAME	CONSTRAINT_NAME	REFERENCED_TABLE_NAME	REFERENCED_COLUMN_NAME
booking	customer_id	fk_booking_customer	customer	customer_id
booking	screening_id	fk_booking_screening	screening	screening_id
booking_seat	booking_id	fk_booking_seat_booking	booking	booking_id
booking_seat	seat_id	fk_booking_seat_seat	seat	seat_id
movie_genre	genre_id	fk_movie_genre_genre	genre	genre_id
movie_genre	movie_id	fk_movie_genre_movie	movie	movie_id
payment	booking_id	fk_payment_booking	booking	booking_id
screen	theater_id	fk_screen_theater	theater	theater_id
screening	movie_id	fk_screening_movie	movie	movie_id
screening	screen_id	fk_screening_screen	screen	screen_id
screening	slot_id	fk_screening_slot	slot	slot_id
seat	screen_id	fk_seat_screen	screen	screen_id
ticket	booking_seat_id	fk_ticket_booking_seat	booking_seat	booking_seat_id

13 rows in set (0.002 sec)

```
mysql>
```

```
mysql> describe movie;
```

Field	Type	Null	Key	Default	Extra
movie_id	int	NO	PRI	NULL	auto_increment
title	varchar(200)	NO		NULL	
duration_minutes	int	NO		NULL	
language	varchar(50)	NO		NULL	
release_date	date	YES		NULL	
certificate	varchar(10)	YES		NULL	
rating	decimal(3,1)	YES		NULL	
synopsis	text	YES		NULL	

```
8 rows in set (0.002 sec)
```

```
mysql> █
```

```
mysql> describe genre;
```

Field	Type	Null	Key	Default	Extra
genre_id	int	NO	PRI	NULL	auto_increment
genre_name	varchar(50)	NO	UNI	NULL	

```
2 rows in set (0.002 sec)
```

```
mysql> █
```

```
mysql> describe theater;
```

Field	Type	Null	Key	Default	Extra
theater_id	int	NO	PRI	NULL	auto_increment
theater_name	varchar(150)	NO		NULL	
city	varchar(100)	NO		NULL	
address	varchar(255)	NO		NULL	
contact_no	varchar(15)	YES		NULL	

```
5 rows in set (0.002 sec)
```

```
mysql> █
```

```
mysql> describe screen;
```

Field	Type	Null	Key	Default	Extra
screen_id	int	NO	PRI	NULL	auto_increment
theater_id	int	NO	MUL	NULL	
screen_name	varchar(100)	NO		NULL	
screen_type	varchar(50)	NO		NULL	
capacity	int	NO		NULL	

```
5 rows in set (0.001 sec)
```

```
mysql> █
```

```
mysql> describe seat;
```

Field	Type	Null	Key	Default	Extra
seat_id	int	NO	PRI	NULL	auto_increment
screen_id	int	NO	MUL	NULL	
row_label	varchar(5)	NO		NULL	
seat_number	int	NO		NULL	
seat_type	varchar(30)	YES		REGULAR	

```
5 rows in set (0.002 sec)
```

```
mysql> █
```

```
mysql> describe slot;
```

Field	Type	Null	Key	Default	Extra
slot_id	int	NO	PRI	NULL	auto_increment
slot_name	varchar(100)	NO	UNI	NULL	
start_time	time	NO		NULL	
end_time	time	NO		NULL	

```
4 rows in set (0.002 sec)
```

```
mysql> █
```

```
mysql> INSERT INTO genre
-> (genre_name)
-> VALUES
-> ('Action'),
-> ('Drama'),
-> ('Sci-Fi'),
-> ('Adventure');
```

Query OK, 4 rows affected (0.001 sec)

Records: 4 Duplicates: 0 Warnings: 0

```
mysql> select * from genre;
```

genre_id	genre_name
7	Action
10	Adventure
8	Drama
9	Sci-Fi

4 rows in set (0.000 sec)

```
mysql> INSERT INTO movie
-> (title, duration_minutes, language, release_date, certificate, rating, synopsis)
-> VALUES
-> ('The Martian', 144, 'English', '2015-10-02', 'UA', 8.0,
->  'An astronaut stranded on Mars uses his scientific knowledge and determination to survive and find a way back home.'),
->
-> ('Project Hail Mary', 150, 'English', '2026-03-20', 'UA', 8.5,
->  'A lone astronaut awakens on a desperate mission to save Earth and uncover the mystery threatening humanity.'),
->
-> ('Gran Turismo', 134, 'English', '2023-08-25', 'UA', 7.1,
->  'A skilled gamer gets the opportunity to turn his virtual racing abilities into a real-world motorsport career.'),
->
-> ('Spider-Man: Brand New Day', 140, 'English', '2026-07-31', 'UA', 8.2,
->  'Spider-Man faces a new chapter of challenges and adventures as he protects his city from new threats.');
```

Query OK, 4 rows affected (0.004 sec)
Records: 4 Duplicates: 0 Warnings: 0

```
mysql> select * from movie;
```

movie_id	title	duration_minutes	language	release_date	certificate	rating	synopsis
1	The Martian	144	English	2015-10-02	UA	8.0	An astronaut stranded on Mars uses his scientific knowledge and determination to survive and find a way back home.
2	Project Hail Mary	150	English	2026-03-20	UA	8.5	A lone astronaut awakens on a desperate mission to save Earth and uncover the mystery threatening humanity.
3	Gran Turismo	134	English	2023-08-25	UA	7.1	A skilled gamer gets the opportunity to turn his virtual racing abilities into a real-world motorsport career.
4	Spider-Man: Brand New Day	140	English	2026-07-31	UA	8.2	Spider-Man faces a new chapter of challenges and adventures as he protects his city from new threats.

4 rows in set (0.000 sec)

```
mysql> █
```

```
mysql> SELECT
->     mg.movie_id,
->     m.title,
->     mg.genre_id,
->     g.genre_name
-> FROM movie_genre mg
-> JOIN movie m
->     ON mg.movie_id = m.movie_id
-> JOIN genre g
->     ON mg.genre_id = g.genre_id
-> ORDER BY mg.movie_id, mg.genre_id;
```

movie_id	title	genre_id	genre_name
1	The Martian	8	Drama
1	The Martian	9	Sci-Fi
2	Project Hail Mary	9	Sci-Fi
2	Project Hail Mary	10	Adventure
3	Gran Turismo	7	Action
3	Gran Turismo	8	Drama
4	Spider-Man: Brand New Day	7	Action
4	Spider-Man: Brand New Day	9	Sci-Fi

8 rows in set (0.000 sec)

```
mysql> █
```

```
mysql> INSERT INTO theater
-> (theater_name, city, address, contact_no)
-> VALUES
-> ('CineSphere Central', 'Hyderabad', 'Banjara Hills Road No. 12', '9876543210'),
-> ('CineSphere Galleria', 'Hyderabad', 'Hitech City Main Road', '9876543211');
```

Query OK, 2 rows affected (0.001 sec)

Records: 2 Duplicates: 0 Warnings: 0

```
mysql> select * from theater;
```

theater_id	theater_name	city	address	contact_no
1	CineSphere Central	Hyderabad	Banjara Hills Road No. 12	9876543210
2	CineSphere Galleria	Hyderabad	Hitech City Main Road	9876543211

2 rows in set (0.000 sec)

```
mysql> █
```

```
mysql> INSERT INTO screen
-> (theater_id, screen_name, screen_type, capacity)
-> VALUES
-> (1, 'Screen 1', 'IMAX', 6),
-> (1, 'Screen 2', 'STANDARD', 6),
-> (2, 'Screen 1', '3D', 6);
```

Query OK, 3 rows affected (0.002 sec)

Records: 3 Duplicates: 0 Warnings: 0

```
mysql> select * from screen;
```

screen_id	theater_id	screen_name	screen_type	capacity
1	1	Screen 1	IMAX	6
2	1	Screen 2	STANDARD	6
3	2	Screen 1	3D	6

3 rows in set (0.001 sec)

```
mysql> █
```

```
mysql> select * from seat;
```

seat_id	screen_id	row_label	seat_number	seat_type
1	1	A	1	REGULAR
2	1	A	2	REGULAR
3	1	A	3	PREMIUM
4	1	B	1	REGULAR
5	1	B	2	REGULAR
6	1	B	3	PREMIUM
7	2	A	1	REGULAR
8	2	A	2	REGULAR
9	2	A	3	PREMIUM
10	2	B	1	REGULAR
11	2	B	2	REGULAR
12	2	B	3	PREMIUM
13	3	A	1	REGULAR
14	3	A	2	REGULAR
15	3	A	3	PREMIUM
16	3	B	1	REGULAR
17	3	B	2	REGULAR
18	3	B	3	PREMIUM

```
18 rows in set (0.000 sec)
```

```
mysql>
```

```
mysql> INSERT INTO slot
-> (slot_name, start_time, end_time)
-> VALUES
-> ('Morning', '10:00:00', '12:30:00'),
-> ('Matinee', '13:00:00', '15:30:00'),
-> ('Evening', '17:00:00', '19:30:00'),
-> ('Night', '20:00:00', '22:30:00');
```

```
Query OK, 4 rows affected (0.002 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql> select * from slot;
```

slot_id	slot_name	start_time	end_time
1	Morning	10:00:00	12:30:00
2	Matinee	13:00:00	15:30:00
3	Evening	17:00:00	19:30:00
4	Night	20:00:00	22:30:00

```
4 rows in set (0.000 sec)
```

```
mysql> █
```

```
mysql> select * from screening;
```

screening_id	movie_id	screen_id	slot_id	screening_date	base_price	status
1	1	1	1	2026-09-10	250.00	SCHEDULED
2	2	1	2	2026-09-10	250.00	SCHEDULED
3	3	2	3	2026-09-10	220.00	SCHEDULED
4	4	3	4	2026-09-10	250.00	SCHEDULED

```
4 rows in set (0.000 sec)
```

```
mysql>
```

```
mysql> SELECT
->     s.screening_id,
->     m.title AS movie,
->     s.screen_id,
->     sl.slot_name,
->     s.screening_date,
->     s.base_price,
->     s.status
-> FROM screening s
-> JOIN movie m
->     ON s.movie_id = m.movie_id
-> JOIN slot sl
->     ON s.slot_id = sl.slot_id
-> ORDER BY s.screening_id;
```

screening_id	movie	screen_id	slot_name	screening_date	base_price	status
1	The Martian	1	Morning	2026-09-10	250.00	SCHEDULED
2	Project Hail Mary	1	Matinee	2026-09-10	250.00	SCHEDULED
3	Gran Turismo	2	Evening	2026-09-10	220.00	SCHEDULED
4	Spider-Man: Brand New Day	3	Night	2026-09-10	250.00	SCHEDULED

4 rows in set (0.001 sec)

```
mysql> █
```

```
mysql> INSERT INTO customer
-> (full_name, email, phone, password_hash)
-> VALUES
-> ('Arjun Mehta', 'arjun@example.com', '90000000001', 'hash_arjun_001'),
-> ('Priya Sharma', 'priya@example.com', '90000000002', 'hash_priya_002'),
-> ('Rahul Verma', 'rahul@example.com', '90000000003', 'hash_rahul_003');
```

Query OK, 3 rows affected (0.001 sec)

Records: 3 Duplicates: 0 Warnings: 0

```
mysql> select * from customer;
```

customer_id	full_name	email	phone	password_hash	registered_at
1	Arjun Mehta	arjun@example.com	90000000001	hash_arjun_001	2026-09-08 20:49:45
2	Priya Sharma	priya@example.com	90000000002	hash_priya_002	2026-09-08 20:49:45
3	Rahul Verma	rahul@example.com	90000000003	hash_rahul_003	2026-09-08 20:49:45

3 rows in set (0.000 sec)

```
mysql>
```

```
mysql> INSERT INTO booking
-> (customer_id, screening_id, total_amount, booking_status)
-> VALUES
-> (1, 1, 500.00, 'CONFIRMED'),
-> (2, 2, 250.00, 'CONFIRMED'),
-> (3, 3, 440.00, 'CONFIRMED'),
-> (1, 4, 250.00, 'CONFIRMED');
```

Query OK, 4 rows affected (0.002 sec)

Records: 4 Duplicates: 0 Warnings: 0

```
mysql> select * from booking;
```

booking_id	customer_id	screening_id	booking_time	total_amount	booking_status
1	1	1	2026-09-08 20:51:14	500.00	CONFIRMED
2	2	2	2026-09-08 20:51:14	250.00	CONFIRMED
3	3	3	2026-09-08 20:51:14	440.00	CONFIRMED
4	1	4	2026-09-08 20:51:14	250.00	CONFIRMED

4 rows in set (0.000 sec)

```
mysql> █
```

```
mysql> SELECT
->     b.booking_id,
->     c.full_name AS customer,
->     m.title AS movie,
->     s.screening_date,
->     sl.slot_name,
->     b.total_amount,
->     b.booking_status
-> FROM booking b
-> JOIN customer c
->     ON b.customer_id = c.customer_id
-> JOIN screening s
->     ON b.screening_id = s.screening_id
-> JOIN movie m
->     ON s.movie_id = m.movie_id
-> JOIN slot sl
->     ON s.slot_id = sl.slot_id
-> ORDER BY b.booking_id;
```

booking_id	customer	movie	screening_date	slot_name	total_amount	booking_status
1	Arjun Mehta	The Martian	2026-09-10	Morning	500.00	CONFIRMED
2	Priya Sharma	Project Hail Mary	2026-09-10	Matinee	250.00	CONFIRMED
3	Rahul Verma	Gran Turismo	2026-09-10	Evening	440.00	CONFIRMED
4	Arjun Mehta	Spider-Man: Brand New Day	2026-09-10	Night	250.00	CONFIRMED

4 rows in set (0.001 sec)

```
mysql> █
```

```
mysql> select * from booking_seat;
```

booking_seat_id	booking_id	seat_id	ticket_price	seat_status
1	1	1	250.00	BOOKED
2	1	2	250.00	BOOKED
3	2	3	250.00	BOOKED
4	3	7	220.00	BOOKED
5	3	8	220.00	BOOKED
6	4	13	250.00	BOOKED

```
6 rows in set (0.000 sec)
```

```
mysql> █
```

```
mysql> select * from ticket;
```

ticket_id	booking_seat_id	issued_at	qr_code	ticket_status
1	1	2026-09-08 20:56:13	QR-CS-0001	ACTIVE
2	2	2026-09-08 20:56:13	QR-CS-0002	ACTIVE
3	3	2026-09-08 20:56:13	QR-CS-0003	ACTIVE
4	4	2026-09-08 20:56:13	QR-CS-0004	ACTIVE
5	5	2026-09-08 20:56:13	QR-CS-0005	ACTIVE
6	6	2026-09-08 20:56:13	QR-CS-0006	ACTIVE

```
6 rows in set (0.001 sec)
```

```
mysql>
```

```
mysql> INSERT INTO payment
-> (booking_id, amount, payment_method, payment_status, transaction_reference)
-> VALUES
-> (1, 500.00, 'UPI', 'SUCCESS', 'TXN-CS-1001'),
-> (2, 250.00, 'CARD', 'SUCCESS', 'TXN-CS-1002'),
-> (3, 440.00, 'NET_BANKING', 'SUCCESS', 'TXN-CS-1003'),
-> (4, 200.00, 'UPI', 'SUCCESS', 'TXN-CS-1004');
```

Query OK, 4 rows affected (0.001 sec)

Records: 4 Duplicates: 0 Warnings: 0

```
mysql> select * from payment;
```

payment_id	booking_id	payment_time	amount	payment_method	payment_status	transaction_reference
1	1	2026-09-08 20:56:47	500.00	UPI	SUCCESS	TXN-CS-1001
2	2	2026-09-08 20:56:47	250.00	CARD	SUCCESS	TXN-CS-1002
3	3	2026-09-08 20:56:47	440.00	NET_BANKING	SUCCESS	TXN-CS-1003
4	4	2026-09-08 20:56:47	200.00	UPI	SUCCESS	TXN-CS-1004

4 rows in set (0.001 sec)

```
mysql> █
```

```
mysql> SELECT * FROM payment;
```

payment_id	booking_id	payment_time	amount	payment_method	payment_status	transaction_reference
1	1	2026-09-08 20:56:47	500.00	UPI	SUCCESS	TXN-CS-1001
2	2	2026-09-08 20:56:47	250.00	CARD	SUCCESS	TXN-CS-1002
3	3	2026-09-08 20:56:47	440.00	NET_BANKING	SUCCESS	TXN-CS-1003
4	4	2026-09-08 20:56:47	250.00	UPI	SUCCESS	TXN-CS-1004

```
4 rows in set (0.000 sec)
```

```
mysql> █
```

```
mysql> SELECT
->     b.booking_id,
->     c.full_name AS customer,
->     m.title AS movie,
->     sc.screen_name,
->     t.theater_name,
->     s.screening_date,
->     sl.slot_name,
->     b.total_amount,
->     b.booking_status
-> FROM booking b
-> JOIN customer c
->     ON b.customer_id = c.customer_id
-> JOIN screening s
->     ON b.screening_id = s.screening_id
-> JOIN movie m
->     ON s.movie_id = m.movie_id
-> JOIN screen sc
->     ON s.screen_id = sc.screen_id
-> JOIN theater t
->     ON sc.theater_id = t.theater_id
-> JOIN slot sl
->     ON s.slot_id = sl.slot_id
-> ORDER BY b.booking_id;
```

booking_id	customer	movie	screen_name	theater_name	screening_date	slot_name	total_amount	booking_status
1	Arjun Mehta	The Martian	Screen 1	CineSphere Central	2026-09-10	Morning	500.00	CONFIRMED
2	Priya Sharma	Project Hail Mary	Screen 1	CineSphere Central	2026-09-10	Matinee	250.00	CONFIRMED
3	Rahul Verma	Gran Turismo	Screen 2	CineSphere Central	2026-09-10	Evening	440.00	CONFIRMED
4	Arjun Mehta	Spider-Man: Brand New Day	Screen 1	CineSphere Galleria	2026-09-10	Night	250.00	CONFIRMED

4 rows in set (0.001 sec)

mysql>

```
mysql> SHOW GRANTS FOR 'role_customer';
```

```
+-----+
| Grants for role_customer@% |
+-----+
| GRANT USAGE ON *.* TO `role_customer`@`%` |
| GRANT SELECT, INSERT ON `cinesphere`.`booking_seat` TO `role_customer`@`%` |
| GRANT SELECT, INSERT ON `cinesphere`.`booking` TO `role_customer`@`%` |
| GRANT SELECT ON `cinesphere`.`genre` TO `role_customer`@`%` |
| GRANT SELECT ON `cinesphere`.`movie` TO `role_customer`@`%` |
| GRANT SELECT, INSERT ON `cinesphere`.`payment` TO `role_customer`@`%` |
| GRANT SELECT ON `cinesphere`.`screen` TO `role_customer`@`%` |
| GRANT SELECT ON `cinesphere`.`screening` TO `role_customer`@`%` |
| GRANT SELECT ON `cinesphere`.`theater` TO `role_customer`@`%` |
| GRANT SELECT ON `cinesphere`.`ticket` TO `role_customer`@`%` |
+-----+
10 rows in set (0.000 sec)
```

```
mysql> SHOW GRANTS FOR 'role_staff';
```

```
+-----+
| Grants for role_staff@% |
+-----+
| GRANT USAGE ON *.* TO `role_staff`@`%` |
| GRANT SELECT, INSERT, UPDATE ON `cinesphere`.`genre` TO `role_staff`@`%` |
| GRANT SELECT, INSERT, UPDATE ON `cinesphere`.`movie` TO `role_staff`@`%` |
| GRANT SELECT, INSERT, UPDATE ON `cinesphere`.`screen` TO `role_staff`@`%` |
| GRANT SELECT, INSERT, UPDATE ON `cinesphere`.`screening` TO `role_staff`@`%` |
| GRANT SELECT, INSERT, UPDATE ON `cinesphere`.`seat` TO `role_staff`@`%` |
| GRANT SELECT, INSERT, UPDATE ON `cinesphere`.`slot` TO `role_staff`@`%` |
| GRANT SELECT, INSERT, UPDATE ON `cinesphere`.`theater` TO `role_staff`@`%` |
+-----+
8 rows in set (0.000 sec)
```

```
mysql> SHOW GRANTS FOR 'role_admin';
```

```
+-----+
| Grants for role_admin@% |
+-----+
| GRANT USAGE ON *.* TO `role_admin`@`%` |
| GRANT ALL PRIVILEGES ON `cinesphere`.* TO `role_admin`@`%` |
+-----+
2 rows in set (0.000 sec)
```

```
mysql>
```